

13.1: First Order Differential Equations:

A differential equation is an equation involving at least one derivative
Order of a differential equation is that of the highest power derivative $\left(\frac{d^n y}{dx^n}\right)$

The solution to a DE containing an arbitrary constant is called a general solution. A particular solution can be found with additional context.

Differentials can be solved by integrating
Model rates of change, for example:

$\frac{d\theta}{dt} = +k(\theta - \theta_0)$, for cooling of an object

13.2: Seperating the variables:

When a differential contains both variables, they must be seperated & only then integrated

$$\frac{dy}{dx} = \frac{f(x)}{g(y)} \Rightarrow g(y)dy = f(x)dx$$
$$\Rightarrow \int g\,dy = \int f\,dx = G(y) = F(x) + C$$